Grading Rubric — Module Two Assignment 2

Electricity Consumption: Preprocessing and Regression

Total: 20 points

How to use this rubric

This rubric is the grading authority. There is no answer key. Each criterion says what the work must *demonstrate*, not what it must say, and several steps have more than one correct implementation.

Four rules apply throughout:

- **Any correct approach earns the marks.** The choice of scaler, encoding, correlation measure, train/test split, degree of polynomial and plotting library are all the student's to make. Do not deduct for choosing differently from another student or from your own preference.
- **"Verify and handle" is satisfied by verifying and finding nothing.** A student who checks for duplicates or missing values, reports that the dataset has none, and moves on has done the step. Requiring them to invent a fix for a problem the data does not have would be wrong.
- **A low accuracy score is not a wrong answer.** Weather variables explain only a small share of power consumption. Grade whether the model was built and evaluated correctly and whether the result was interpreted honestly — never the size of R^2.
- **Code must run.** A cell that was never executed, or that ends in a traceback, has not produced the result it claims. Judge the code on its merits and say so in the comment, but do not credit an output that does not exist.

Step 1. Data preprocessing — 5 points

The assignment states that neglecting preprocessing costs marks, so this is scored across its five named requirements rather than as one judgment.

- **1 pt — Duplicated rows.** Checks for repeated rows (e.g. `duplicated()`) and either removes them or states that none were found.
- **1 pt — Missing values.** Checks for missing values (e.g. `isnull().sum()`) and either handles them by a stated method or states that none were found.
- **1 pt — Categorical to numerical.** Handles `Date Time`, which is the only non-numeric column. Any defensible treatment earns the mark: parsing it and extracting components (hour, month, day of week), converting to a numeric timestamp, or dropping it with a stated reason. Silently leaving a string column in the feature matrix does not.
- **1 pt — Feature scaling.** Applies a scaler to the numeric predictors, or argues that scaling is unnecessary for ordinary least squares. Both earn the mark; doing neither does not.
- **1 pt — Feature selection.** Selects features, or states on the evidence of the correlation analysis that all predictors are retained. The requirement says "if necessary", so a reasoned decision either way is full credit; silence is not.

Award each of the five independently. A student who does four of the five scores 4.

Step 2. Multicollinearity — 3 points

- **2 pts** — **Analysis.** Produces a correlation matrix over the predictors AND a visualization of it (a heatmap, or a pair plot / scatter matrix). Award 1 if the matrix is computed but never visualized, or visualized but never computed.
- **1 pt** — **Interpretation.** Names which variables are collinear with which, from their own output, and says what that implies for regression (unstable or uninterpretable coefficients, redundant predictors). A plot with no reading of it earns 0 here.

A student who additionally computes VIF has gone beyond the requirement. That is welcome but earns no extra marks, and its absence costs none.

Step 3. Simple linear regression — 4 points

The assignment requires **two** separate single-feature models, then a comparison.

- **1 pt** — **Feature choice.** Identifies two features as the "best" linear predictors AND ties the choice to the correlation analysis from Step 2. Any pair that follows from their own numbers is correct. A pair chosen with no stated reason earns 0.
- **2 pts** — **Two models built and evaluated.** Fits a separate simple linear regression on each of the two features and reports an accuracy score (R^2, or another metric the student names) for each. Award 1 if only one model was built, or if two were built but only one was scored.
- **1 pt** — **Visualization and comparison.** Plots the fit for the feature(s) — a scatter of the data with the regression line is the expected form — and states which of the two scored better.

Step 4. Polynomial regression — 3 points

- **2 pts** — **Built and evaluated.** Fits a polynomial regression, states the degree used, and reports an accuracy score. Award 1 if the model is fitted but not scored, or scored but the degree is never stated.
- **1 pt** — **Visualization.** Plots the polynomial fit against the data.

Any degree is acceptable. A student who tries several degrees and reports how the score changed has done more than was asked, which is welcome but not required.

Step 5. Multiple linear regression — 3 points

- **2 pts** — **Built and evaluated.** Fits a regression on multiple predictors and reports an accuracy score.
- **1 pt** — **Visualization.** Any defensible way of visualizing a multi-predictor fit earns the mark: predicted against actual, a residual plot, or a bar chart of coefficients. Do not require a form the assignment did not name.

Step 5a. Comparison of the three models — 2 points

- **2 pts** — Compares simple, polynomial and multiple regression against each other using **their own reported scores**, and explains *why* the results came out in that order — for example that multiple regression uses more information than a single feature, or that a polynomial fits curvature a straight line cannot, or that a higher degree began to overfit.
- **1 pt** — Lists the three scores side by side with no explanation of the difference, or explains a difference that their own numbers do not show.
- **0** — No comparison, or a comparison of fewer than three models.

The conclusion is not fixed. Whichever model came out ahead in the student's own run is the right answer for them, provided the explanation matches their numbers.

Cross-cutting deductions

These apply on top of the criteria above and are the only grounds for deducting outside them:

- **Explanations.** The submission requirements ask for thorough explanations of graphs rather than one-liners. Where a criterion above awards a mark for interpretation, a single unexplained sentence does not earn it.
- **Code that does not run.** If a cell was never executed or ended in an error, the marks that depend on its output are not earned. Marks for code that is correct but unexecuted may still be awarded where the criterion is about method rather than result — say which applies in the comment.

What the data does and does not fix

There is no answer key for this assignment and no single correct implementation. Grade against the rubric criteria.

What is fixed by the data: the target variable is **Power consumption**; **Date Time** is the only non-numeric column, so it is the one that must be converted or dropped; the five remaining predictors are Temperature, Humidity, Wind Speed, General diffuse flows and Diffuse flows.

What is **not** fixed, and must be judged from the student's own output rather than assumed: whether this copy of the file contains duplicated rows or missing values at all, and which features turn out to be the two "best" for simple regression. A student who checks for duplicates and missing values, reports that there are none, and moves on has completed that requirement — the requirement is to verify and handle, and "none found" is a valid outcome of verifying. Likewise, the two features chosen for simple regression need only follow from the correlation analysis the student actually performed.

Accuracy on this dataset is expected to be poor for simple regression — weather variables explain only a small share of power consumption, and a low R^2 that the student reports honestly and explains is a correct result, not a failed one. Do not reward a high score or penalise a low one; grade the method and the interpretation.